ICSFF: Information Constraint on Self-Supervised Feature Fusion for Few-Shot Remote Sensing Image Classification

Chen Yang, *Member, IEEE*, Tongtong Liu, Guanhua Chen, and Wenhui Li

***Abstract*— The self-supervised few-shot remote sensing image classification task is to achieve efficient and accurate remote sensing image classification through autonomous learning and feature exploitation with limited data labels. However, at this stage, one common challenge in self-supervised learning is the significant disparity between the self-supervised learning task and the main classification task. This disparity can lead to a situation where the model overly emphasizes features or local information emphasized by the self-supervised task while neglecting the essential global semantic information relevant to the main classification task. To solve these problems, this article proposes a few-shot remote sensing image classification framework based on information constraint on self-supervised feature fusion (ICSFF). First, we train a supervised model to capture important semantic information. Then, we leverage this supervised information to constrain the learning process of the self-supervised model. We utilize an attention mechanism to integrate supervised information and self-supervised information through a graph structural feature fusion approach, resulting in enhanced feature representations. In addition, we design a new feature extractor called GCCANet. It helps the model to better utilize the key features by incorporating the global attention module, the group convolution, the residual operation, and the channel shuffle module techniques. We conduct comparative experiments on three public remote sensing datasets, and the experimental results show that ICSFF achieves outstanding performance in the remote sensing image classification task.**

***Index Terms*— Feature fusion, few-shot learning, information constraint, remote sensing image classification, self-supervised learning.**

I. Introduction

REMOTE sensing technology encompasses the utilization of remote sensing platforms, such as satellites and airplanes, for collecting valuable data about the Earth's surface, atmosphere, and oceans. Its applications span across diverse fields, including agriculture, forestry, water resources, meteorology, geological exploration, urban planning, and environmental protection [1], [2], [3], [4], [5].

Remote sensing image classification is the automatic identification and classification of surface objects using images of the surface of the Earth obtained by remote sensing technology. Nevertheless, traditional approaches to remote sensing image classification encounter challenges in handling few-shot problems. Given the diverse and variable nature of objects on the Earth's surface, obtaining a substantial amount of labeled data becomes difficult, considering the vast volume of image data available [6].

The few-shot remote sensing image classification methods refer to techniques employed for classifying remote sensing images using a limited amount of labeled data [7], [8], [9], [10], [11], [12], [13], [14], [15]. These methods require fewer labeled samples compared to traditional approaches while maintaining a high level of classification accuracy. Typically, few-shot remote sensing image classification methods leverage techniques such as transfer learning and meta-learning. By incorporating relevant domain-specific prior knowledge and utilizing existing data and models, these methods enable efficient training and classification [16]. In light of this, prototype network methods have been introduced [17], incorporating the notion of averaging examples from each class within the learning representation space of a neural network. By evaluating the distance to the prototype representation of each class in the learning metric space, classification can be conducted. This facilitates the model's ability to rapidly learn and classify new classes, even in scenarios with limited labeled data.

Since the introduction of self-supervised learning for few-shot remote sensing image classification, it has greatly reduced the reliance on costly manual labeling and made model training for remote sensing image classification tasks more feasible and economical. This is because self-supervised learning methods can be pre-trained with large amounts of unlabeled data by designing pretextual tasks and make the representation learned by the model have better generalization ability [18], [19], [20]. In addition, the application of self-supervised learning allows for the extension of the training data with unlabeled samples, thus enhancing the ability of the framework to feature extraction from remote sensing images without adding additional requirements [21]. Based on the advantages of self-supervised learning, Li et al. [21] proposed SCL-MLNet, which incorporates self-supervised comparative learning into few-shot learning. This enables the feature extraction networks to learn informative image features even with limited data.

However, incorporating self-supervised learning tasks into few-shot classification scenarios is challenging due to the limited availability of labeled data. This scarcity can lead to a phenomenon called "supervision collapse" [22], where the model overemphasizes capturing low-level features like

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The authors are with the College of Computer Science and Technology, Jilin University, Changchun 130012, China (e-mail: liwh@jlu.edu.cn).

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color histograms and edge continuity, rather than focusing on essential semantic information. This bias in the self-supervised learning task can misdirect the primary classification task, ultimately resulting in a degradation of classification performance. Moreover, existing methods for self-supervised few-shot remote sensing image classification predominantly focus on the differences between samples from base classes, neglecting the valuable semantic information inherent within these samples [23]. Consequently, these methods may struggle to effectively utilize the limited sample data to extract additional semantic information when encountering samples from new classes. This limitation significantly impacts the characterization ability and leads to degraded classification performance. Hence, extracting more semantic information becomes crucial in order to obtain enhanced representations when tackling few-shot classification tasks.

In this article, we propose an information constraint on self-supervised feature fusion (ICSFF) for few-shot remote sensing image classification framework. Specifically, in the ICSFF framework, we initially leverage supervised learning to learn representative supervised information. Subsequently, we use this supervised information as a constraint to guide the learning process of the self-supervised model. ICSFF employs a graph structural feature fusion approach with an attention mechanism to integrate the supervised information and self-supervised information into a unified distribution. This integration enhances and broadens the original representation. In addition, we design a new feature extractor to better capture the high-level features of the data by changing the structure to focus on both global and local information of the remote sensing images. We conduct experiments on three public remote sensing datasets, and the experimental results demonstrate that the ICSFF framework outperforms other general few-shot learning methods and state-of-the-art self-supervised few-shot learning methods.

The main contributions of this article are as follows.

1) We introduce a novel framework for few-shot remote sensing image classification, namely ICSFF. In contrast to semi-supervised approaches, our framework does not rely on any auxiliary unlabeled data.

2) We employ the information obtained through supervised learning as a constraint to optimize the learning process of the self-supervised model. This constraint aids in shaping and enhancing the feature manifold of self-supervised learning, leading to a reduction in representation bias and enabling the extraction of more effective semantic information.

3) We utilize a novel feature fusion approach based on an attention mechanism and graph structure. This approach facilitates the integration of features into a unified distribution, resulting in an enhanced representation of features.

4) We develop a novel backbone architecture that incorporates parallel scanning of large and small convolutional kernels. This design allows the model to effectively capture both global and local information from remote sensing images.

The rest of the article is organized as follows. Section II reviews related work on traditional remote sensing image classification, few-shot remote sensing image classification, and graph convolutional neural networks (GCNs). Section III details our proposed ICSFF framework and the new feature extractor designed. Section IV presents the experimental results and analysis of three public remote sensing datasets as well as further discussion on the effectiveness of the methodology. Section V summarizes the article and gives conclusions.

II. Related Works

In this section, we mainly introduce related work on traditional remote sensing image classification, few-shot remote sensing image classification and graph convolutional networks (GCNs).

A. Traditional Remote Sensing Image Classification

Deep learning has emerged as a powerful technique in machine learning algorithms, allowing the extraction of more informative features through the utilization of massive training data and deep models with multiple hidden layers. Chen et al. [24] proposed a CNN approach for feature extraction and classification of hyperspectral images. Their method employed a deep network architecture consisting of multiple convolutional and pooling layers, with a softmax classifier used for classification. In addition, the vision transformer (ViT), introduced by Dosovitskiy et al. [25], has achieved impressive results across various computer vision tasks by effectively capturing global interactions through learning relationships between sequence elements. Building upon this, Bazi et al. [26] proposed a remote sensing image classification model called RS-ViT. RS-ViT utilizes the ViT pre-training model for feature extraction, and classification is performed using fully connected layers and the softmax function. However, it is important to note that these methods often rely on a significant amount of labeled training data to achieve optimal results. Applying deep learning to remote sensing datasets with a limited number of labeled samples may result in overfitting and may not be universally applicable for remote sensing image classification.

B. Few-Shot Remote Sensing Image Classification

Due to the small number of remote sensing datasets compared to traditional image classification datasets, the problem of remote sensing image classification is usually handled by few-shot learning. In few-shot learning, various strategies are usually employed to deal with the problem of insufficient training examples. For example, data augmentation techniques are used to enhance the dataset, migration learning is applied by using pretrained models for remote sensing image classification tasks, or meta-learning is employed to quickly learn and adapt to new datasets with limited instances. First, Vinyals et al. [27] proposed matching networks. The main idea of the method is to use the information in the small label support set to infer the labels of the unlabeled samples and to achieve label mapping by similarity matching. Finn et al. [28] proposed a model-agnostic meta-learning algorithm that is compatible with any model trained by gradient descent. The algorithm utilizes a small amount of training data from a new task for a small amount of gradient training to achieve good generalization performance on that task. Li et al. [29] proposed Meta-SGD, which is a stochastic gradient descent meta-learning algorithm that is easy to train and requires

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only one step to initialize and tune any differentiable learner. In addition, Sung et al. [30] proposed a relation network that can be trained end to end. During the meta-learning process, each cluster in the relation network is designed for a small number of samples. Once the training is complete, it can classify images of new classes by calculating the relation scores between the query image and a small number of samples for each new class without further updating the network. Li et al. [31] proposed discriminative learning for adaptive matching networks, which first uses an attention technique to mine the relationship between channels and space to automatically discover discriminative regions. Considering the large intraclass variability and interclass similarity, the authors spliced discriminative features from the support and query collections along the depth, utilized a matcher to adaptively select semantically related sample pairs, and computed similarity scores. As the quality of remote sensing images continues to improve, Cheng et al. [32] noticed the issue that most prototype-based few-shot methods directly calculate prototypes on the support sample set and ignore the validation of prototypes. This leads to a decrease in accuracy when reasoning based on prototypes. To address the above issues, the authors proposed a Siamese Prototype Network with Prototype Self-calibration and Intercalibration. Self-calibration refers to calibrating the prototype of the support features using the supervisory information provided by the labels of the support set, and intercalibration refers to using the confidence scores of the samples in the query set as another form of prototype to predict the support set samples in the same way. After that, Chen et al. [33] proposed a deep nearest-neighbor neural network based on an attentional mechanism, which employs a channel attentional mechanism with global information to design an attentional map related to the scene classes to suppress the influence of irrelevant regions. It then uses the attention mechanism to weigh the similarity between the query image and the class image to get the metric score between the image and the class.

Training with a small number of samples, compared to a large number of samples, can result in lower classification accuracy, primarily due to potential biases in the dataset distribution. As a result, the model may struggle to classify certain classes effectively. The limited amount of data makes it challenging for the model to adequately learn and understand the unique characteristics and variations associated with each class. Furthermore, few-shot remote sensing image classification methods may be unable to fully exploit deeper information and patterns present in large-scale datasets. The restricted number of samples limits the model's capacity to capture complex patterns and subtle differences accurately.

C. Graph Convolutional Networks

GCN [34] is a neural network specialized in processing graph data that performs convolutional operations on graphs to extract effective feature representations. The basic idea of GCN is to transform the nodes and edges on the graph into a matrix representation and then implement the convolution operation through matrix operations. The convolution operation here is defined based on the multiplication operation of graph adjacency matrices and can be understood as a special kind of convolution operation performed on graphs. In GCN, the feature representation of each node is jointly determined by its features as well as the features of its neighboring nodes. The final feature representation of each node can be obtained by continuously iterating this process of feature transfer. Suppose there is a batch of graph data with N nodes, each node has its features, we set the features of these nodes to form an $N \times D$ dimensional matrix X, and then the relationship of each node will also form an $N \times N$ dimensional adjacency matrix A. Then, matrix X and matrix A are the inputs to our model. GCN is also a neural network layer that propagates from layer to layer as shown below

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right) \tag{1}$$

where $\tilde{A} = A + I$, I is the unit matrix. $\tilde{D}$ is the degree matrix of $\tilde{A}$. σ is a nonlinear activation function.

The advantage of GCN is that it can handle non-euclidean structured data and has good interpretability. In addition, GCN can be easily combined with other deep learning models to further improve the expressive power of the model.

III. Method

A. Overview

In order to learn more accurate feature representations in few-shot remote sensing image classification tasks, we propose an information constraint on self-supervised feature fusion for few-shot remote sensing image classification called ICSFF framework. As shown in Fig. 1, we begin by employing a supervised learning method to train an initial feature extractor, denoted as $f_\varphi(\cdot)$. Subsequently, we utilize the SimSiam [35] approach to train a self-supervised feature extractor, denoted as $g_\omega(\cdot)$, leveraging the learned supervisory information to guide the self-supervised learning stage. Finally, we introduce a novel feature fusion technique that combines the supervised and self-supervised information. This fusion method incorporates attention mechanisms and graph structures to embed the final features into the network $h_\theta(\cdot)$. We exclusively rely on the D_{train} dataset for training, which fundamentally sets it apart from semisupervised learning methods.

B. Preliminary

In data preparation for few-shot learning, the dataset is usually divided into two parts: the support set and the query set. The support set S is used for model training and contains a small number of samples and corresponding labels. The query set Q is used for model evaluation and prediction and contains a large number of samples and corresponding labels. We divide each dataset D into a training set D_{train}, a validation set D_{val}, and a test set D_{test}. The above division has its own labeling space and ensures that each set does not intersect with each other, then we have $L_{\text{train}} \cap L_{\text{val}} = \emptyset$, $L_{\text{train}} \cap L_{\text{test}} = \emptyset$, $L_{\text{val}} \cap L_{\text{test}} = \emptyset$, and $|L_{\text{total}}| = |L_{\text{train}} \cup L_{\text{val}} \cup L_{\text{test}}|$. In order to train and test using a meta-learning approach, we divide the few-shot task into several subtasks T. The goal of each subtask T_i is to use the support set to estimate the classes of the samples in the query set. The support set S_i for each task contains K different classes with C samples in each

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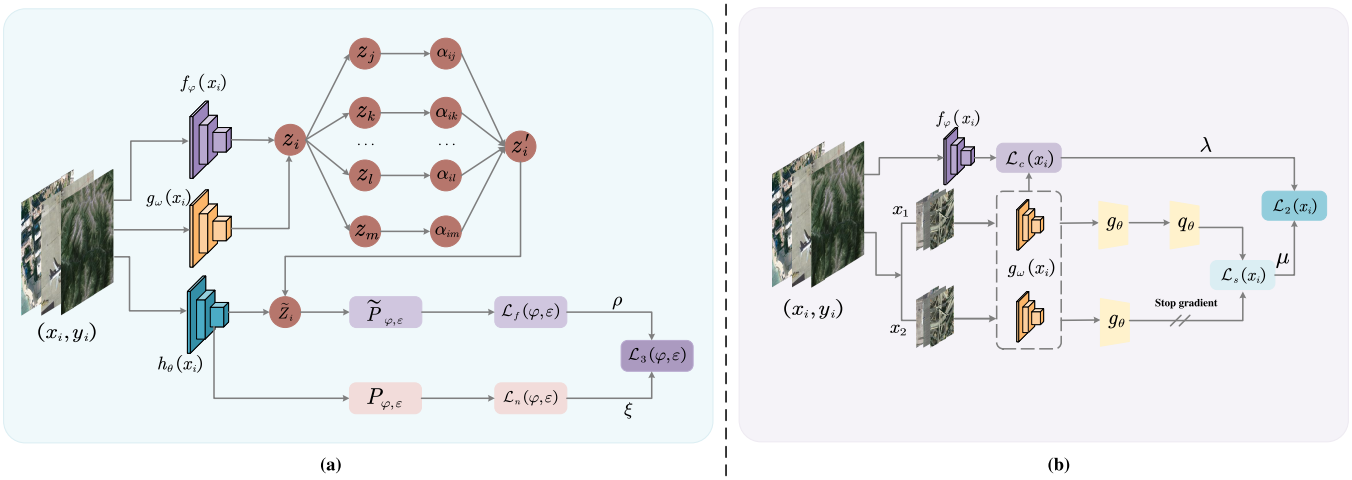


Fig. 1. Illustration of the proposed ICSFF framework describes the process of (a) feature fusion and (b) self-supervised learning with information constraints. The final training model $g_\omega(\cdot)$ from stage (b) is used in stage (a).

class. This is the few-shot classical problem K-way C-shot problem. Similarly for the query, set Q_i contains K different classes with N samples in each class. As a result, there are $D_{\text{train}} = \{T_i\} = \{S_i, Q_i\}$, $D_{\text{val}} = \{T_x\} = \{S_x, Q_x\}$, and $D_{\text{test}} = \{T_y\} = \{S_y, Q_y\}$, where T_x and T_y are different subtasks, S_x and S_y are support sets for the corresponding tasks, and Q_x and Q_y are query sets for the corresponding tasks.

Both the number of classes and the number of samples we apply in our experiments depend on the actual needs. The five-way one-shot and five-way five-shot evaluation criteria are usually used in remote sensing image classification tasks.

C. Information Constraint

As shown in Fig. 1(b), we mainly determine the prototypes of each set of training samples by calculating the similarity between the samples, and then use the prototypes in the classification task of the test samples. Assuming that the mean of the training sample vectors of each class is set to m, then the center of mass of the training samples of class k is denoted as

$$m_k = \frac{1}{|S_k|} \sum_{(x_i, y_i) \in S_k} f_\varphi(x_i) \tag{2}$$

where S_k is the support set sample labeled as class k. In the testing stage, for each test sample, the similarity between it and the prototypes of all classes is computed and the test sample is assigned to the class with which it has the highest similarity. In this article, we choose the cosine similarity as the distance function $d_\varepsilon(\cdot)$, and in order to produce a distribution among all classes, we select a query sample x_q from the query set Q_i, which has the distribution expression

$$p_{\varphi,\varepsilon}\left(y = k | x_q\right) = \frac{\exp\left(-d_\varepsilon\left(f_\varphi\left(x_q\right), m_k\right)\right)}{\sum_{k'=1}^{c} \exp\left(-d_\varepsilon\left(f_\varphi\left(x_q\right), m_{k'}\right)\right)} \tag{3}$$

where $d_\varepsilon(A, B) = \lambda \cos(F_\varepsilon(A), F_\varepsilon(B))$, λ is the inverse parameter of temperature, and $F_\varepsilon(\cdot)$ is a single-layer neural network parameterized by ε. We minimize the loss function by updating the parameters φ and ε through continuous training, and the model reaches a converged state. The expression of the supervised information loss function for ICSFF is shown in the following equation:

$$\mathcal{L}_1(\varphi, \varepsilon) = \underset{(x_q, y_q) \in Q_i}{E} \left[-\log P_{\varphi,\varepsilon}\left(y = k | x_q\right)\right]. \tag{4}$$

The key to the self-supervised training process is to utilize the natural structure or prior knowledge in the data to generate labels and use the generated labels to train the model. A self-supervised feature extractor needs to be trained first, and the self-supervised model we have chosen is the SimSiam method. As shown in Fig. 1(b), we give an input image x and create two different augmented views x_1 and x_2 of the same image, passing the augmented views to the feature extractor $g_\omega(\cdot)$, which generates two feature vectors, one for each augmented view. Next, we pass the feature vectors to the projection head $g_\theta(\cdot)$, which is a small neural network that maps features into a low-dimensional space. This produces two projection vectors. One of the projection vectors is then passed to the prediction head $q_\theta(\cdot)$ to generate a prediction vector, which is another small neural network that maps the vector back to the same dimension of the original feature space. The goal of SimSiam is to minimize the distance between the prediction vector and the other projection vector, so we give the expression for the prediction vector as

$$p_1 = q_\theta(g_\theta(g_\omega(x_1))). \tag{5}$$

The expression for the projection vector is shown in the following equation:

$$z_{2=} g_\theta(g_\omega(x_2)). \tag{6}$$

So, the distance between the prediction vector and the projection vector is shown in the following equation:

$$D(p_1, z_2) = -\frac{p_1}{\|p_1\|_2} \cdot \frac{z_2}{\|z_2\|_2} \tag{7}$$

where $\|\cdot\|_2$ is the l_2-norm. Furthermore, the most important part of the method is the realization of the stop-gradient (stopgrad(·)) [36]. Since the same operation is applied to both augmented views, we perform the stop-gradient operation on z_2 and z_1, respectively. The expression of the loss function

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in the self-supervised learning is shown in the following equation:

$$\mathcal{L}_s(x) = \frac{1}{2}D(p_1, \text{stopgrad}(z_2)) + \frac{1}{2}D(p_2, \text{stopgrad}(z_1)). \tag{8}$$

We utilize supervised information as a constraint to optimize the features learned by the self-supervised module, thereby reducing the occurrence of biased representative features learned by the self-supervised model. The ICSFF framework utilizes the negative cosine similarity operation between supervised and self-supervised information as a constraining condition. The expression of its loss function is shown below

$$\mathcal{L}_c(x) = -\frac{f_\varphi(x)}{\|f_\varphi(x)\|_2} \cdot \frac{g_\omega(x)}{\|g_\omega(x)\|_2}. \tag{9}$$

So, the total loss function for the self-supervised learning can be defined as

$$\mathcal{L}_2(x) = \underset{x \in D_{\text{train}}}{E} [\mu\mathcal{L}_s(x) + \lambda\mathcal{L}_c(x)] \tag{10}$$

where μ and λ are importance coefficients weighing the self-supervised symmetric loss and the constraining condition.

D. Feature Fusion

As shown in Fig. 1(a), the ICSFF framework introduces a graph structure feature fusion approach based on attention mechanism. This method effectively leverages information from multiple feature graphs to enhance the model's expressive capacity. By incorporating attention mechanisms, the framework can selectively focus on relevant features and exploit their synergistic effects, leading to improved model performance. Specifically, we give a sample x_i and compute its corresponding two embedding vectors $f_\varphi(x_i)$ and $g_\omega(x_i)$. We use the incremental operation $z_i = \text{concat}(f_\varphi(x_i), g_\omega(x_i))$ splice to derive the eigenvector of z_i. We construct a nearest neighbor graph $G(V, E)$ based on the cosine similarity [37], and the similarity expression between samples and samples is shown in the following equation:

$$\cos(z_i, z_j) = \frac{z_i^T z_j}{\|z_i\|_2 \|z_j\|_2} \tag{11}$$

where V is the vertex matrix, each vertex v_i represents a feature of a sample, and E is the edge of the graph structure denoting the similarity between vertices.

In the aggregation process, we assume that the central node is v_i and the weight of the attention mechanism [38] from the neighbor node v_j to v_i is assumed to be

$$e_{ij} = \sigma\left[w^T \cdot \frac{1}{2}(Wz_i + Wz_j)\right] \tag{12}$$

where w is the weight parameter, σ is the activation function, and W is the weight matrix of the node feature transformation, which is shared by all weights.

In order to better assign the weights, we need to normalize the correlation calculated with all neighbors in a uniform way to get the weight coefficients as shown in the following equation:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{v_k \in N(v_i)} \exp(e_{ik})}. \tag{13}$$

Thus, once we have obtained the weight coefficients, we can follow the idea of weighted summation by the attention mechanism. The new eigenvector of node v_i is shown in the following equation:

$$z_i' = \sigma\left(\sum_{v_j \in N(v_i)} \alpha_{ij} W z_j\right). \tag{14}$$

Next, we can generate a new feature matrix with weights $z' = (z_1', z_2', \ldots, z_n')$, where n is the number of samples. We construct an adjacency matrix A using cosine similarity and compute the new feature vector similarity expression as

$$\cos(z_i', z_j') = \frac{z_i', : {z_j'}^T, :}{\|z_i', : \|_2 \|z_j', : \|_2} \tag{15}$$

where : denotes taking all the elements and $\|\cdot\|$ denotes the mode of the vector.

Each vertex represents a sample feature $A_{i,j}$ with the following expression:

$$A_{i,j} = \begin{cases} \cos(z_i', z_j'), & i \neq j \\ 0, & i = j. \end{cases} \tag{16}$$

We use E as the Laplace matrix in the GCN to aggregate the information between vertices with the expression shown in the following equation:

$$E = D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \tag{17}$$

where D is the degree diagonal matrix, $D_{ii} = \Sigma_j A_{i,j}$.

For sample x_i, the eigenvector expression of x_i is obtained by distillation [37] as shown in the following equation:

$$\tilde{Z}_i = \Sigma_{j=1}^{n} (\beta I + E)_{i,j}^{v} h_\theta(x_j) \tag{18}$$

where I is the unit matrix, v is the number of aggregated features, and β is a balance parameter used to weigh the representation of a neighbor against its own representation.

We select a query sample x_q from the query set Q_i, then its distribution expression is shown in the following equation:

$$\widetilde{P}_{\varphi,\varepsilon}(y = k|x_q) = \frac{\exp(-d_\varepsilon(\tilde{Z}_i, \tilde{m}_k))}{\sum_{k'=1}^{c} \exp(-d_\varepsilon(\tilde{Z}_i, \tilde{m}_{k'}))} \tag{19}$$

where the center of mass of the k class of training samples is denoted

$$\tilde{m}_k = \frac{1}{|S_k|} \sum_{(x_i, y_i) \in S_k} \tilde{Z}_i. \tag{20}$$

If we do not use the feature fusion method, we also select a query sample x_q from the query set Q_i with the distribution expression shown in the following equation:

$$P_{\varphi,\varepsilon}(y = k|x_q) = \frac{\exp(-d_\varepsilon(h_\theta(x_q), m'_k))}{\sum_{k'=1}^{c} \exp(-d_\varepsilon(h_\theta(x_q), m'_{k'}))} \tag{21}$$

where the center of mass of the k class of training samples is denoted

$$m'_k = \frac{1}{|S_k|} \sum_{(x_i, y_i) \in S_k} h_\theta(x_q). \tag{22}$$

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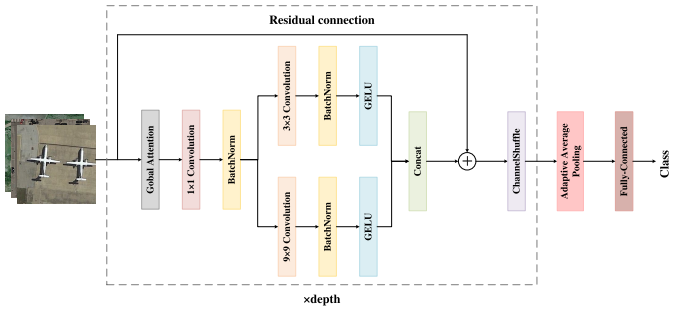


Fig. 2. General architecture of the feature extractor GCCANet.

Therefore, we give the final loss function expression as

$$\mathcal{L}_3(\varphi, \varepsilon) = \underset{(x_q, y_q)\epsilon\, Q_i}{E} \left[-\rho \log \widetilde{P}_{\varphi,\varepsilon}\left(y = k|x_q\right) - \xi P_{\varphi,\varepsilon}\left(y = k|x_q\right) \right] \tag{23}$$

where ρ and ξ are balance coefficients, which are used to weigh the significance between the two losses.

E. Feature Extractor

The backbone feature extraction network is to extract the key features from the input image which can be used to classify the image properly. As shown in Fig. 2, we design a new grouped convolutional channel attention network GCCANet based on the convmixer network [39]. First, we employ a global attention module based on the channel attention mechanism, which is a module for computing the global attention representation of the input tensor. In a CNN, each channel corresponds to a class of features. In some cases, the features corresponding to certain channels contribute more to the task, while the features corresponding to other channels are relatively less important [40]. Therefore, the role of the global attention module based on the channel attention mechanism is to compute the contribution of each channel to the task so that the model can better utilize the features in the input tensor. Specifically, the module first reduces the number of channels of the input tensor to 1 using a 1×1 convolutional layer, and then computes the importance of each channel for the global features by means of a softmax operation and restores it to its original shape. It results in a weight tensor that represents the contribution of each channel to the task. This weight tensor can be multiplied with the input tensor to obtain a feature representation that enhances the important channel features.

Next, we employ the ideas of group convolution [41] and tensor splicing. Specifically, the structure divides the number of channels of the input tensor into groups. A convolution operation is performed for each group using a different convolution kernel, and then the output tensor of each group is spliced in the channel dimension to obtain the final output tensor. It is possible to further increase the depth and width of the network while ensuring the efficiency and expressiveness of the model, thus improving the performance of the model. In addition, we also introduce the residual operation.

Then, we introduce the channel shuffle module [42]. Specifically, we first use a convolutional layer to reduce the number of input channels to a specified multiple, then use the channel shuffle module to exchange and utilize the information between different groups, and finally use another convolutional layer to restore the number of channels to their original size. This process helps the model to make better use of the information between different groups, thus improving the performance and generalization of the model.

IV. Experiments

In this section, our primary objective is to showcase the effectiveness of our method through a series of experiments. First, we conduct a comprehensive comparison between our method and both state-of-the-art and classical methods on three publicly available datasets. This allows us to evaluate the performance of our approach in relation to existing solutions. Second, we perform experiments to empirically determine the optimal hyperparameters for different loss functions utilized in our method. This careful analysis ensures that our model is fine-tuned for optimal performance. Furthermore, we compare our proposed GCCANet with three classical networks, providing a thorough evaluation of its superiority in terms of various performance metrics. In addition, we conduct visualization experiments to gain insights into the inner workings of our model and demonstrate its ability to capture meaningful patterns and relationships in the data. At last, we conduct ablation studies to further validate the performance of our model.

A. Dataset Description

We conduct experiments on three public remote sensing datasets, mainly including the Northwestern Polytechnical University (NWPU)-RESISC45 dataset [43], the aerial image dataset (AID) dataset [44], and the Wuhan University (WHU)-RS19 dataset [45], and for effective experimental comparisons, we follow the division of the training set, the validation set, and the test set of the other comparison experiments [21], [31].

NWPU-RESISC45 dataset is a large-scale public dataset published by Northwestern Polytechnic University, which contains 45 classes of scenes. Each class contains 700 remote sensing images, and the whole dataset contains 31 500 images, each of which is 256×256 in size. The entire dataset is randomly selected from 25 classes for the training dataset, including airplane, baseball diamond, beach, bridge, chaparral, church, cloud, desert, freeway, golf course, harbor, island, lake, meadow, mobile home park, mountain, palace, railway, rectangular farmland, roundabout, sea ice, ship, sparse residential, stadium, and wetland. It is randomly selected from ten classes for the validation dataset, including commercial area, industrial area, overpass, railway station, runway, snowberg, storage tank, tennis court, terrace, and thermal power station. It is randomly selected from ten classes for the test dataset, including airport, basketball court, circular farmland, dense residential, forest, ground track field, intersection, medium residential, parking lot, and river.

The AID dataset is a large AID that collects sample images from Google Earth images. It contains 30 scene classes, with roughly 220–420 images per class. The entire dataset has 10 000 images, each with a pixel size of 600×600 pixels.

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The entire dataset is selected from 16 classes for the training dataset, including bare land, baseball field, bridge, church, commercial, dense residential, farm land, forest, industrial, medium residential, mountain, parking, pond, port, resort, and school. It is selected from seven classes for the validation dataset, including beach, center, meadow, park, playground, square, and stadium. It is selected from seven classes for the test dataset, including airport, desert, railway station, river, sparse residential, storage tanks, and viaduct.

The WHU-RS19 dataset is released by the Wuhan University. It is a set of satellite images exported from Google Earth, which can provide high-resolution satellite images up to 0.5 m. It contains 19 different scene classes, and the sample size of each class is greater than or equal to 50 images. The entire dataset consists of 1005 images, each with a pixel size of 600 × 600 pixels. Nine classes are randomly selected for the training dataset, including airport, bridge, desert, football field, industrial, mountain, parking, port, and residential. It is randomly selected from five classes for the validation dataset, including beach, farmland, forest, park, and railway station. It is randomly selected from five classes for the test dataset, including commercial, meadow, pond, river, and viaduct.

B. Experimental Setting

We implement our model using the PyTorch development framework, which is deployed on an NVIDIA GeForce RTX 3090 GPU with CUDA version 11.4. We use the Adam optimizer for optimization with the learning rate set to 0.001.

We set the size of all remote sensing image datasets to 128 × 128, and the random data augmentation methods used are random horizontal flipping, random cropping, color jittering, and random grayscale transformation. Then 100 episodes are randomly generated for training of remote sensing images, 500 episodes for validation of remote sensing images, and 2000 episodes for testing of remote sensing images. In each episode, the K-way C-shot problem is followed. We randomly select five classes of images in each episode for both the support set and query set, and each class contains one or five samples in the experiments. For comparison with other experiments, we give 15 sample examples for each class of the query set.

C. Overall Experimental Results and Comparisons

For the comparison of our experiments, we select some classical and state-of-the-art few-shot remote sensing image classification methods, such as the conventional metric learning methods ProtoNet [17], MatchingNet [27], RelationNet [30], and SPNet [32], gradient descent methods such as MAML [28] and Meta-SGD [29], self-supervised learning methods such as SCL-MLNet [21], attention mechanism methods such as DLA-MatchNet [31] and DN4AM [33], as well as multimodal learning methods such as TeAw [46]. For a fair comparison, the feature extractors used in our experiments are ResNet12. In addition, we also conduct experiments using the GCCANet as the feature extractor. Our experiments strictly adhere to the model evaluation metrics for the K-way C-shot problem, and all experiments in the testing stage take the average accuracy of 2000 episodes to compute the overall accuracy, which serves as a final measure of the effectiveness of the model.

TABLE I
FEW-SHOT CLASSIFICATION ACCURACY COMPARISON ON THE NWPU DATASET

Method	Backbone	5-way 1-shot	5-way 5-shot
ProtoNet [17]	ResNet12	61.69±0.33%	81.28±0.49%
MatchingNet [27]	ResNet12	64.46±0.28%	80.83±0.69%
MAML [28]	ResNet12	62.28±0.46%	79.82±0.73%
Meta-SGD [29]	ResNet12	65.51±0.78%	81.36±0.53%
RelationNet [30]	ResNet12	66.69±0.72%	82.10±0.88%
DLA-MatchNet [31]	ResNet12	69.32±0.35%	83.26±0.34%
SCL-MLNet [21]	ResNet12	67.01±0.44%	83.57±0.26%
SPNet [32]	ResNet12	68.42±0.55%	83.98±0.66%
DN4AM [33]	ResNet12	70.15±0.82%	85.69±0.69%
TeAw [46]	ResNet12	70.23±0.45%	85.77±0.25%
ICSFF	ResNet12	70.27±0.49%	86.16±0.56%
ICSFF	GCCANet	**70.93±0.53%**	**86.83±0.77%**

TABLE II
FEW-SHOT CLASSIFICATION ACCURACY COMPARISON ON THE AID DATASET

Method	Backbone	5-way 1-shot	5-way 5-shot
ProtoNet [17]	ResNet12	61.26±0.62%	81.66±0.29%
MatchingNet [27]	ResNet12	62.68±0.34%	79.32±0.81%
MAML [28]	ResNet12	62.44±0.55%	78.26±0.83%
Meta-SGD [29]	ResNet12	63.78±0.44%	80.22±0.72%
RelationNet [30]	ResNet12	66.27±0.13%	84.37±0.43%
DLA-MatchNet [31]	ResNet12	66.15±0.21%	85.74±0.56%
SCL-MLNet [21]	ResNet12	65.29±0.43%	83.85±0.66%
SPNet [32]	ResNet12	64.17±0.52%	82.39±0.39%
DN4AM [33]	ResNet12	67.82±0.89%	86.03±0.42%
TeAw [46]	ResNet12	68.13±0.36%	86.47±0.25%
ICSFF	ResNet12	68.27±0.49%	87.16±0.56%
ICSFF	GCCANet	**68.75±0.54%**	**87.73±0.27%**

We select the metrics for the final test results to be consistent with those selected by most methods at this stage for five-way one-shot and five-way five-shot experiments. As shown in Table I, we conduct experiments on the NWPU dataset and the experimental results can show that our proposed framework ICSFF has the highest results, reaching 70.93% in the five-way one-shot result, outperforming the second place TeAw by 0.70%. In the five-way five-shot result, it reaches 86.83%, outperforming the second place TeAw by 1.06%. Table II shows our experimental results on the AID dataset. According to the experimental results in the table, it can be seen that our method outperforms all other remote sensing image classification methods in terms of classification results. The accuracy of the model can reach 68.75% in the five-way one-shot result and 87.73% in the five-way five-shot result. Our experimental results on the WHU dataset are shown in Table III. The WHU dataset has a relatively small amount of data and reaches 87.31% in the five-way one-shot result, which is 1.99% higher than the TeAw method which ranks second. In the five-way five-shot result, the accuracy of the model can reach 97.56%, which is able to outperform some traditional methods by more than 10%.

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TABLE III
FEW-SHOT CLASSIFICATION ACCURACY COMPARISON ON THE WHU DATASET

Method	Backbone	5-way 1-shot	5-way 5-shot
ProtoNet [17]	ResNet12	61.29±0.31%	82.03±0.16%
MatchingNet [27]	ResNet12	61.36±0.24%	82.82±0.82%
MAML [28]	ResNet12	53.04±0.64%	79.02±0.53%
Meta-SGD [29]	ResNet12	55.92±1.29%	65.77±0.68%
RelationNet [30]	ResNet12	69.41±0.99%	85.34±0.97%
DLA-MatchNet [31]	ResNet12	74.17±0.60%	87.03±0.33%
SCL-MLNet [21]	ResNet12	70.52±0.47%	86.61±0.62%
SPNet [32]	ResNet12	70.48±0.83%	85.45±0.43%
DN4AM [33]	ResNet12	84.98±0.63%	96.72±0.20%
TeAw [46]	ResNet12	85.32±0.52%	96.97±0.16%
ICSFF	ResNet12	85.27±0.49%	97.16±0.56%
ICSFF	GCCANet	**87.31±0.25%**	**97.56±0.08%**

Overall, our introduced ICSFF framework performs well in remote sensing image classification. By analyzing the experimental results, we can conclude that our approach is more suitable for datasets with fewer and fewer sample types. Both for the five-way one-shot task and the five-way five-shot task, our proposed model can achieve the highest accuracy, which indicates that our method can effectively target the few-shot remote sensing image classification task.

D. Feature Extractor Comparison Experiments

In order to understand the impact of different feature extractors on the few-shot remote sensing image classification task, we investigate the performance of different feature extractors based on the ICSFF framework. As shown in Figs. 3 and 4, we select three networks to compare with our proposed GCCANet, which are ConV4, ResNet12 [47], and ConvMixer, the first two networks are the classical networks for few-shot remote sensing image classification. From the experimental results in the figure, it can be seen that ConV4 has the lowest classification accuracy on three datasets and GCCANet has the highest classification accuracy on three datasets, both in the five-way one-shot task and the five-way five-shot task. On the NWPU dataset, the classification accuracy of ResNet12 is somewhat higher than that of ConvMixer. On the AID dataset and WHU dataset, the classification accuracies of ResNet12 and ConvMixer are not much different from each other. Therefore, GCCANet combines techniques such as global attention module, group convolution, residual operation, and channel shuffle module, aiming to improve the performance and generalization ability of the feature extraction network. The combination of these techniques can help the model to better utilize key features and improve image classification performance.

E. Comparative Experiments on Feature Fusion

In order to further validate the effectiveness of the feature fusion approach for graph structures based on the attention mechanism proposed by the ICSFF framework, we conduct comparative experiments on feature fusion on three datasets.

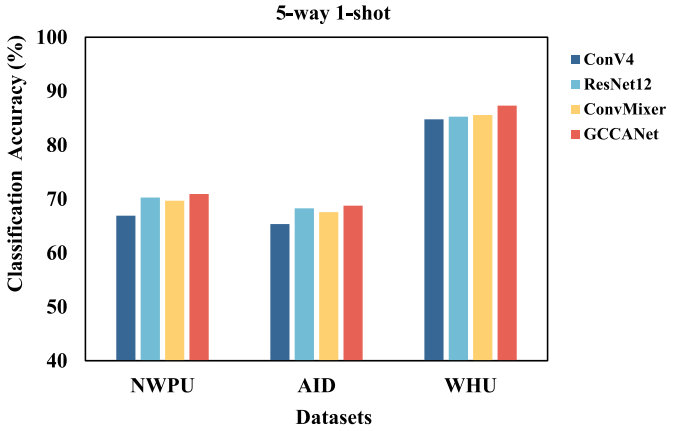


Fig. 3. Five-way one-shot classification accuracy for four different feature extractors on three datasets.

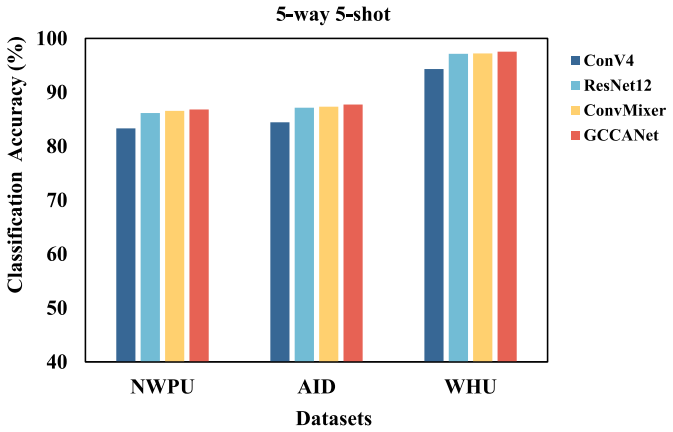


Fig. 4. Five-way five-shot classification accuracy for four different feature extractors on three datasets.

We define the approach without feature fusion as NFFD and the feature fusion approach for graph structures without attention mechanism as GCN. As shown in Table IV, without the use of feature fusion, both the five-way one-shot results and the five-way five-shot results have the lowest classification accuracies, which can be about 4% lower on all three datasets. On the NWPU dataset, for the five-way one-shot task, the classification accuracy of the feature fusion approach without using the graph structure based on the attention mechanism can reach 68.34%, but it can still differ by 2.59% compared to our proposed method. Similarly, for the five-way five-shot task, the accuracy of the former can reach 85.07%, which is 1.76% lower compared to our proposed method. On the AID dataset, for the five-way one-shot task, the classification accuracy of our proposed method is 2.49% higher than that of the feature fusion approach that does not use the graph structure based on the attention mechanism. Similarly, for the five-way five-shot task, the former has 1.84% higher accuracy than the latter. On the WHU dataset, for the five-way one-shot task, the classification accuracy of the feature fusion approach without using the graph structure based on the attention mechanism can reach 87.31%. For the five-way five-shot task, the classification accuracy of the feature fusion approach using the graph structure based on the attention mechanism can reach 97.56%. From the results, we use the attention mechanism to allow the model to automatically learn how much attention to pay to different features, which enables the model to dynamically adjust the weights of the

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TABLE IV
FEATURE FUSION COMPARISON EXPERIMENTS ON THREE DATASETS

Method	NWPU		AID		WHU	
	5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
NFFD	66.65±0.57%	82.71±0.45%	65.03±0.78%	83.31±0.62%	83.58±0.49%	93.02±0.17%
GCN	68.34±0.64%	85.07±0.29%	66.26±0.44%	85.89±0.38%	85.60±0.37%	95.97±0.09%
ICSFF	**70.93±0.53%**	**86.83±0.77%**	**68.75±0.54%**	**87.73±0.27%**	**87.31±0.25%**	**97.56±0.08%**

features, so as to choose to pay attention to the important features and suppress the unimportant ones. This feature fusion approach improves the ability of the model to perceive critical information and enhances the classification performance.

F. Visualization Experiments

In order to demonstrate more intuitively the impact of the graph structure feature fusion approach of the attention mechanism on the remote sensing classification task, we plot the class heat maps by the XGrad-CAM method [48]. As shown in Fig. 5, we randomly select three classes for our experiments, the first column represents the original input image, the second column represents the heat map without the feature fusion method, the third column represents the heat map without an attention mechanism for the feature fusion method, and the fourth column represents the heat map of the ICSFF framework. From the results in the figure, we can see that for airplane identification, the heat map shows that our model focuses on the wing-to-fuselage joint of the airplane, which indicates that this region is very critical for distinguishing airplanes. Whereas it can be seen from the heat map without using the feature fusion approach that only the approximate positions of the two airplanes can be identified, the heat map without using the attention mechanism approach shows that the identification of part of the effective key features is lost. For the identification of commercial areas, our model focuses on the protruding part of the building, which indicates that it is very crucial for the identification of commercial areas. Heat maps that do not use the feature fusion method are less effective in extracting key parts, while heat maps that do not use the attention mechanism method are able to roughly localize the key locations, but still lose some of the key features. For the recognition of tennis courts, it can be seen from the heat map that our model focuses on the overall shape of the tennis court. While a large amount of invalid information in the heat map without using the feature fusion method is also regarded as a high contributing feature, there is still a small amount of noise in the heat map without using the attention mechanism method is regarded as a high contributing feature.

With these heat maps, we can visually observe that the model gives higher attentional weights on key regions for different classes of objects. This proves the effectiveness of the ICSFF framework in feature extraction, which is able to accurately capture the important features that distinguish the classes.

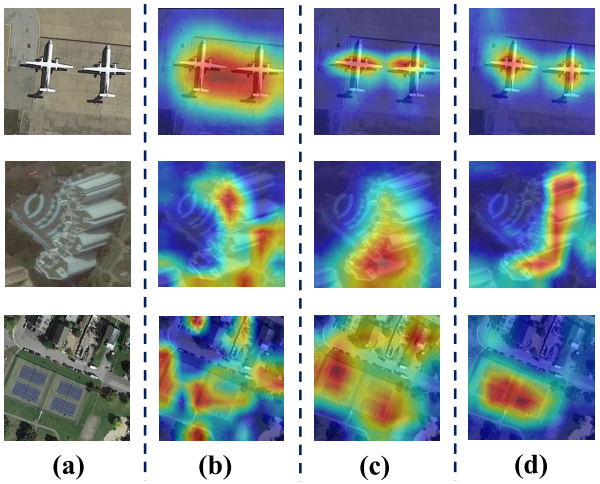


Fig. 5. Comparison of heat maps for three classes. The class in the first row is airplanes, the class in the second row is commercial areas, and the class in the third row is tennis courts. The red areas are high contribution areas. The blue areas are low contribution areas. (a) Original input image, (b) heat map without the feature fusion method, (c) heat map without an attention mechanism for the feature fusion method, and (d) heat map of the ICSFF framework.

G. Selection of Parameters

In this section, we discuss some hyperparameters and weighting coefficients in order to obtain the optimal ICSFF framework. First, in the information constraint stage, in order to obtain the optimal loss function, we conduct multiple experiments of weighting coefficient selection on three public remote sensing datasets. As shown in Fig. 6, the horizontal axis represents the loss ratio μ/λ of the self-supervised symmetric loss to the constraining condition, and the vertical axis represents the classification accuracy. We choose 12 sets of values for different cases, the first nine sets where the sum of μ and λ is equal to 1, and the last three sets where we ensure that μ is equal to 1 and adjust the value of λ. In the figure, we can see that for the first nine datasets, as the value of μ keeps decreasing by 0.1 and the value of λ keeps increasing by 0.1, the classification accuracies of the five-way one-shot and the five-way five-shot on both the NWPU dataset and the AID dataset keep decreasing. For the WHU dataset, as the value of μ keeps decreasing by 0.1 and the value of λ keeps increasing by 0.1, the fold line tends to decrease to 0.8/0.2, then increase to 0.6/0.4, and then decrease to 0.1/0.9. For the last three datasets, we can see from the graphs that there is a rise and then a decrease in all three datasets, with all fold lines peaking at $\mu = 1$, $\lambda = 0.5$, and above the maximum in the first nine datasets. From the results, it can be seen that the self-supervised symmetric loss and the constraining conditional loss account for a larger proportion of the self-supervised symmetric loss, and the appropriate introduction of the constraining conditional loss can effectively improve the accuracy of the model. Therefore, we choose

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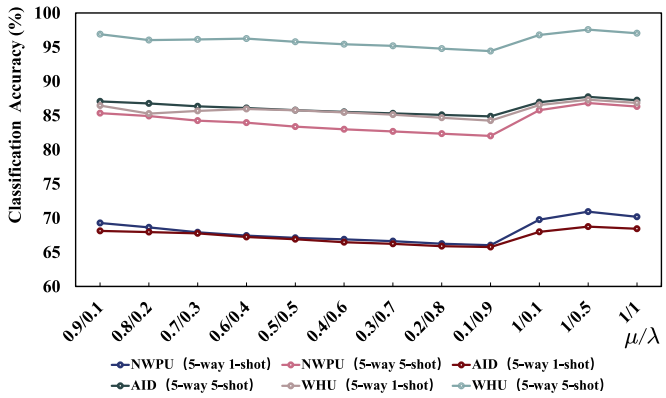


Fig. 6. Comparative experiments on the selection of weighting coefficients on three datasets during the information constraint stage.

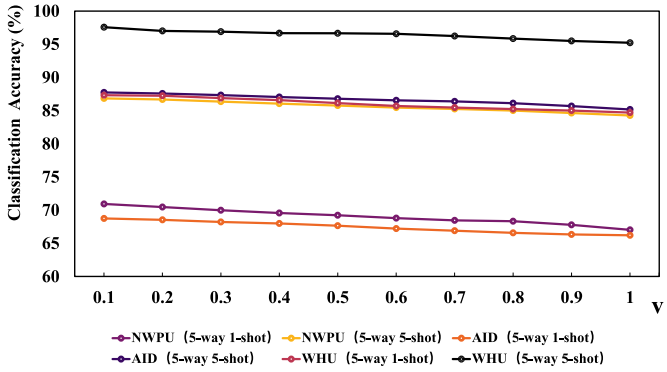


Fig. 7. Comparative experiments for selecting the number of aggregated features on three datasets during the feature fusion stage.

1 as the coefficient for the self-supervised symmetric loss and 0.5 as the coefficient for the constraining conditional loss.

The second hyperparameter we discuss is the number of times v that features are aggregated when performing a feature aggregation operation. As shown in Fig. 7, the horizontal axis represents the value of v and the vertical axis represents the classification accuracy. We conduct ten sets of experiments for different values of the number of aggregated features. Typically, the values of v range from 0–1. Based on the trend of the fold line in the figure, it can be seen that as the value of v continues to increase, the classification accuracy of the model decreases both five-way one-shot and five-way five-shot, so that it is able to reach its peak when the value of v is set to 0.1. According to the experimental results, we believe that if the value of v is too large, the information of the feature vectors may be overaggregated, which leads to the difficulty of the model to capture the subtle differences between different nodes, thus affecting the performance and generalization ability of the model. Therefore, we choose the number of aggregated features to be 0.1, which can avoid the node features being overaggregated, reduce the loss of information, and retain more local information.

Finally, we want to discuss the choice of weighting coefficients in the loss function in the feature fusion stage. As shown in Fig. 8, the horizontal axis represents the ratio of loss using feature fusion to loss without feature fusion ρ/ξ, and the vertical axis represents the classification accuracy. We take the same values for the weighting coefficients in Fig. 8 as we do in Fig. 7, choosing a total of 12 sets of values for the

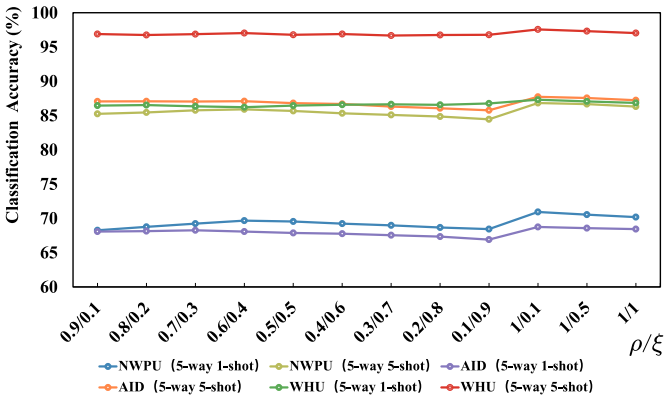


Fig. 8. Comparison experiments of weighting coefficients of loss ratios using feature fusion versus loss ratios without feature fusion are selected on three datasets during the feature fusion stage.

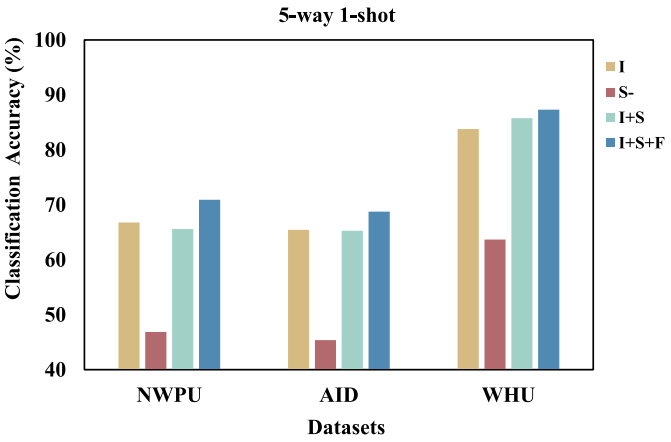


Fig. 9. Five-way one-shot classification accuracy for ablation experiments on three datasets.

different cases, with the first nine sets of the sum of ρ and ξ equal to 1, and the last three sets ensuring that ρ is equal to 1 and adjusting the value of ξ. From the results in the figure, we can see that on the NWPU dataset, the trend of the fold lines becomes increasing for ξ values of 0.1–0.4, and the classification accuracy of the model keeps decreasing as the ξ values range from 0.5 to 0.9. For the last three datasets, the classification accuracy of the model is able to peak when ξ is equal to 0.1. On the AID dataset, for the five-way five-shot results, the folds remain essentially stable for ρ values of 0.6–0.9. As ρ decreases from 0.6 all the way to 0.1, the classification accuracy of the model decreases. For the five-way one-shot results, the trend of the fold lines is increasing for values of ρ from 0.7 to 0.9. As ρ decreases from 0.7 to 0.1, the classification accuracy of the model decreases. For the last three datasets, the classification accuracy of the model is able to peak when ξ is equal to 0.1. On the WHU dataset, the trend of the fold lines remains essentially stable and constant as ξ continues to increase from 0.1. For the last three datasets, the same results appear as for the other two datasets, where the classification accuracy of the model is able to peak when ξ is equal to 0.1. From the results, it can be seen that the loss of using feature fusion accounts for a larger proportion of the loss of using feature fusion than the loss of not using feature fusion. Therefore, we choose 1 as the coefficient of loss using

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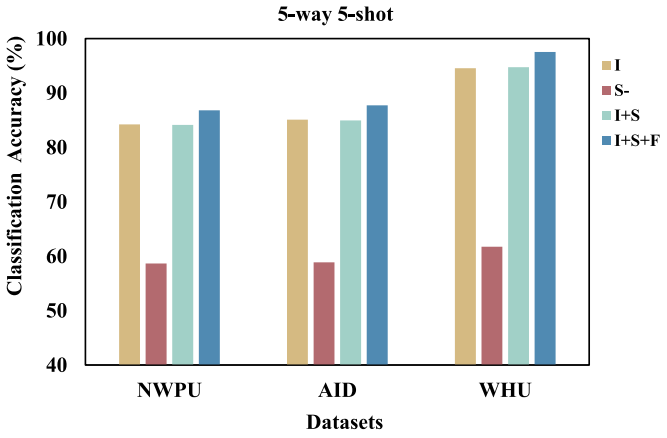


Fig. 10. Five-way five-shot classification accuracy for ablation experiments on three datasets.

feature fusion and 0.1 as the coefficient of loss without feature fusion.

H. Ablation Experiments

To examine the effects of information constraint and feature fusion on remote sensing image classification tasks, we conduct ablation experiments. In the first experiment, denoted as I, we solely rely on supervised information and employ the f_φ as the feature extractor for classification. In the second experiment, denoted as $S-$, we exclude the information constraint during the self-supervised learning stage. In the third experiment, denoted as $I + S$, we combine supervised information with information constraint self-supervised learning, utilizing the g_ω as the feature extractor for classification. Finally, in the fourth experiment, denoted as $I + S + F$, we incorporate supervised information, information constraint self-supervised learning, and feature fusion, employing the h_θ as the feature extractor for classification. As shown in Figs. 9 and 10, it is evident that the classification accuracy significantly decreases when the information constraint is not applied during self-supervised learning. This effectively demonstrates the challenge of learning meaningful representations through self-supervised learning with limited labeled data. On the NWPU and AID datasets, the classification accuracy of using only supervised information is slightly higher than that of combining supervised information with information constraint self-supervised learning. However, on the WHU dataset, incorporating information constraint self-supervised learning leads to slightly higher classification accuracy compared to using only supervised information. Nevertheless, the difference in performance between these two sets of experiments is not substantial. This effectively demonstrates that the addition of the information constraint enables effective representation learning through self-supervised learning. Furthermore, the model's classification accuracy is significantly improved when feature fusion is applied. This experimental result highlights the efficacy of our proposed ICSFF framework in enhancing model performance and addressing the challenges of few-shot remote sensing image classification tasks.

V. Conclusion

In this article, we propose a self-supervised few-shot remote sensing image classification framework called ICSFF, which enhances model performance through feature fusion. The method utilizes supervised information as an information constraint during self-supervised learning. Subsequently, a graph structure feature fusion method, incorporating attention mechanisms, is employed to enhance the representation of features and improve model performance. In addition, we improve the feature extractor, GCCANet fuses the global attention module, group convolution, residual operation, and channel shuffle module techniques, which enables the model to better utilize the key features and improve the accuracy of image classification. We validate the model on three public remote sensing datasets, and the experimental results demonstrate the effectiveness of our proposed ICSFF framework.

However, unsupervised learning still faces challenges in remote sensing image classification at the current stage. As self-supervised learning gradually shows promising results in remote sensing image classification, we expect future models to contribute to unlabeled datasets.

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Chen Yang (Member, IEEE) received the B.S. degree from the Changchun University of Science and Technology, Changchun, Jilin, China, in 2021. She is currently pursuing the M.S. degree with the College of Computer Science and Technology, Jilin University, Changchun.

Her research interests include computer vision, self-supervised learning, and few-shot learning.



Tongtong Liu received the B.S. degree in software engineering from Southwest Jiaotong University, Chengdu, China, in 2017. She is currently taking successive postgraduate and doctoral programs of study for Ph.D. degree with the College of Computer Science and Technology, Jilin University, Changchun, Jilin, China.

Her research interests include computer vision, multilabel classification, and zero-shot learning.



Guanhua Chen received the B.S. degree from the Changchun University of Science and Technology, Changchun, Jilin, China, in 2022. He is currently pursuing the M.S. degree with the College of Computer Science and Technology, Jilin University, Changchun.

His research interests include image processing and computer vision.



Wenhui Li received the B.S. and M.S. degrees in computer science and the Ph.D. degree from Jilin University, Changchun, Jilin, China, in 1984, 1987, and 1996, respectively.

His major research interests include computer vision, image processing, pattern recognition, graphics, virtual reality, CAD, and geometric constraint.

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